

BEATRICE, ON, Canada

WMO: 716940

Lat: 45.1361N

Lon: 79.3989W

Elev: 297

StdP: 97.81

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.0	-24.4	-31.2	0.2	-27.4	-27.7	0.3	-24.2	7.0	-6.2	6.3	-9.2	1.1	80	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	28.6	20.9	27.2	20.1	25.8	19.2	22.4	26.4	21.5	25.1	20.6	24.0	3.1	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.2	16.4	24.3	20.2	15.5	23.3	19.4	14.6	22.5	67.4	26.4	63.8	25.0	60.7	23.9	27.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	6.2	5.4	4.8	DB	-34.4	31.1	3.1	1.2	-36.6	32.0	-38.5	32.7	-40.2	33.4	-42.5	34.3
(5)				WB	-34.5	24.3	3.1	1.3	-36.7	25.3	-38.6	26.0	-40.3	26.8	-42.6	27.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-9.3	-9.1	-3.4	4.1	11.5	15.8	18.6	17.5	13.6	7.5	0.8	-5.5	(6)
(7)		DBStd	11.25	7.72	6.81	6.85	4.92	4.70	3.44	2.80	2.92	4.23	4.48	5.33	6.43	(7)
(8)		HDD10.0	2653	599	536	417	184	38	2	0	0	13	105	277	481	(8)
(9)		HDD18.3	4865	858	769	673	426	217	90	31	50	151	335	526	739	(9)
(10)		CDD10.0	926	0	0	2	9	85	177	268	233	122	28	1	0	(10)
(11)		CDD18.3	97	0	0	0	0	7	15	40	25	10	0	0	0	(11)
(12)		CDH23.3	1032	0	0	2	6	97	168	426	247	85	2	0	0	(12)
(13)		CDH26.7	171	0	0	0	0	18	26	77	37	12	0	0	0	(13)
(14)	Wind	WSAvg	2.0	2.1	2.2	2.3	2.4	2.1	1.8	1.8	1.7	1.7	2.0	2.0	2.1	(14)
(15)	Precipitation	PrecAvg	1031	104	72	68	65	82	73	91	74	99	104	106	99	(15)
(16)		PrecMax	1201	163	115	139	121	134	101	163	115	141	178	171	148	(16)
(17)		PrecMin	846	60	22	20	21	27	33	25	23	20	24	58	36	(17)
(18)		PrecStd	87	28	26	34	26	30	20	39	25	31	43	32	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.3	7.9	18.8	24.0	28.8	29.0	30.6	30.0	28.5	23.4	17.2	9.7	(19)
(20)			MCWB	7.9	6.1	13.9	14.5	20.1	20.8	21.6	20.6	21.8	16.5	12.2	9.1	(20)
(21)		2%	DB	5.1	4.6	11.5	19.4	26.1	27.2	28.7	27.5	25.9	20.1	13.0	6.3	(21)
(22)			MCWB	4.4	3.4	7.0	10.8	18.5	19.7	21.4	20.4	20.2	15.9	9.9	5.5	(22)
(23)		5%	DB	2.8	2.0	8.4	15.9	23.6	25.3	27.2	26.0	23.6	17.7	10.9	4.4	(23)
(24)			MCWB	2.0	0.6	5.1	9.6	15.8	18.2	20.2	19.7	18.5	14.5	8.4	3.4	(24)
(25)		10%	DB	0.9	0.6	5.7	13.3	21.1	23.5	25.7	24.5	21.8	15.3	8.7	2.3	(25)
(26)			MCWB	0.3	-0.5	3.0	8.1	14.0	17.0	19.2	18.8	17.6	12.8	7.1	1.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	6.7	14.4	15.4	21.3	23.0	24.1	23.2	22.5	19.3	13.3	9.4	(27)
(28)			MCDWB	8.3	6.8	19.3	19.6	27.2	27.5	28.5	27.0	26.7	20.5	15.7	9.9	(28)
(29)		2%	WB	4.4	3.5	8.3	13.2	19.5	21.2	22.5	22.0	21.2	16.6	10.6	5.6	(29)
(30)			MCDWB	4.8	4.6	10.5	17.3	24.5	25.0	26.5	25.2	24.4	18.8	12.5	6.2	(30)
(31)		5%	WB	2.2	0.9	5.3	10.7	17.6	19.8	21.5	20.9	19.9	14.9	8.8	3.5	(31)
(32)			MCDWB	2.8	1.6	7.8	14.4	21.6	23.0	25.2	24.0	22.4	17.4	10.3	4.2	(32)
(33)		10%	WB	0.3	-0.4	3.3	8.7	15.8	18.6	20.5	19.9	18.5	13.1	7.1	1.6	(33)
(34)			MCDWB	0.8	0.5	5.5	12.8	19.1	21.7	24.1	23.1	20.8	14.8	8.6	2.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.6	11.2	11.3	11.5	13.4	12.2	12.4	12.1	12.2	9.8	8.6	8.1	(35)
(36)			MCDBR	8.3	9.8	13.4	17.6	16.7	15.0	14.6	14.0	14.2	13.9	12.6	7.8	(36)
(37)			MCWBR	7.9	8.8	9.3	10.4	8.9	8.3	8.0	7.8	8.7	9.1	9.7	7.2	(37)
(38)		5% WB	MCDBR	8.3	9.4	11.5	14.7	13.4	11.9	11.8	10.9	11.6	11.5	10.9	7.5	(38)
(39)			MCWBR	8.2	9.2	8.9	10.0	8.0	7.5	7.1	6.8	7.8	8.5	9.2	7.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.288	0.309	0.349	0.372	0.399	0.402	0.394	0.368	0.340	0.306	0.279	(40)	
(41)		taud	2.376	2.344	2.414	2.392	2.373	2.316	2.331	2.344	2.410	2.480	2.519	2.442	(41)	
(42)		Ebn at Noon	833	899	928	912	896	870	862	856	851	829	807	804	(42)	
(43)		Edh at Noon	74	92	100	112	118	126	122	116	100	82	66	63	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.32	2.21	3.60	4.49	5.26	5.74	5.80	5.00	3.91	2.30	1.34	0.98	(44)	
(45)		RadStd	0.16	0.28	0.37	0.41	0.51	0.45	0.41	0.28	0.38	0.25	0.17	0.12	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page